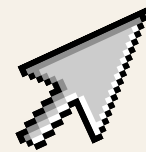


LIFT ELECTRONIC CONTROLLER SYSTEM PROJECT



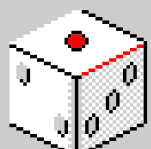
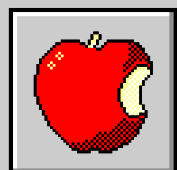
DIGITAL LOGIC
(SECR1013-03)

Lecturer: Dr. Nurfaizrina binti Mohd Zamry

Group 5 Members



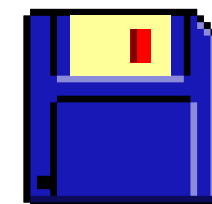
TAN ZHI MING	A23CS0189
NG YU HIN	A23CS0148
LING YU QIAN	A23CS0301
KOO XUAN	A23CS0300



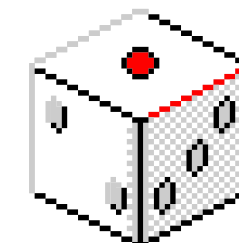
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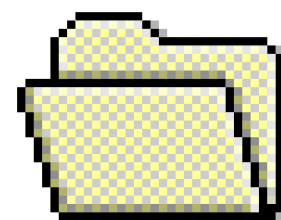
CONTENTS



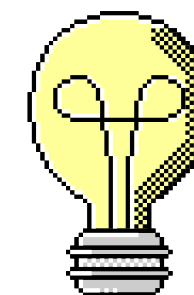
The Background



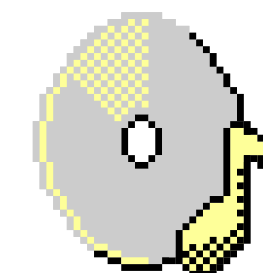
Problem & Solution



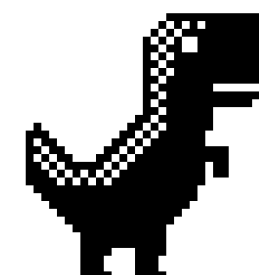
Block Diagram



The Requirements



System Implementation



Full Deeds Drawing



Conclusion

Start

The Background

Components of

Combinational Circuit

- Switches
- Decoder
- Basic gates
- LEDs
- 3-bit comparator
- 7-segment display

Sequential Circuit

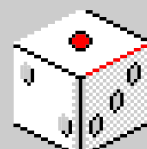
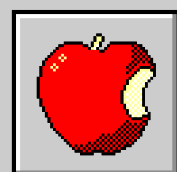
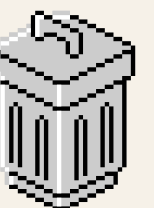
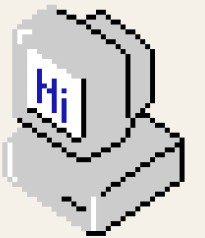
- 3-bit up/down counter
- Clock disabler

Enhanced features

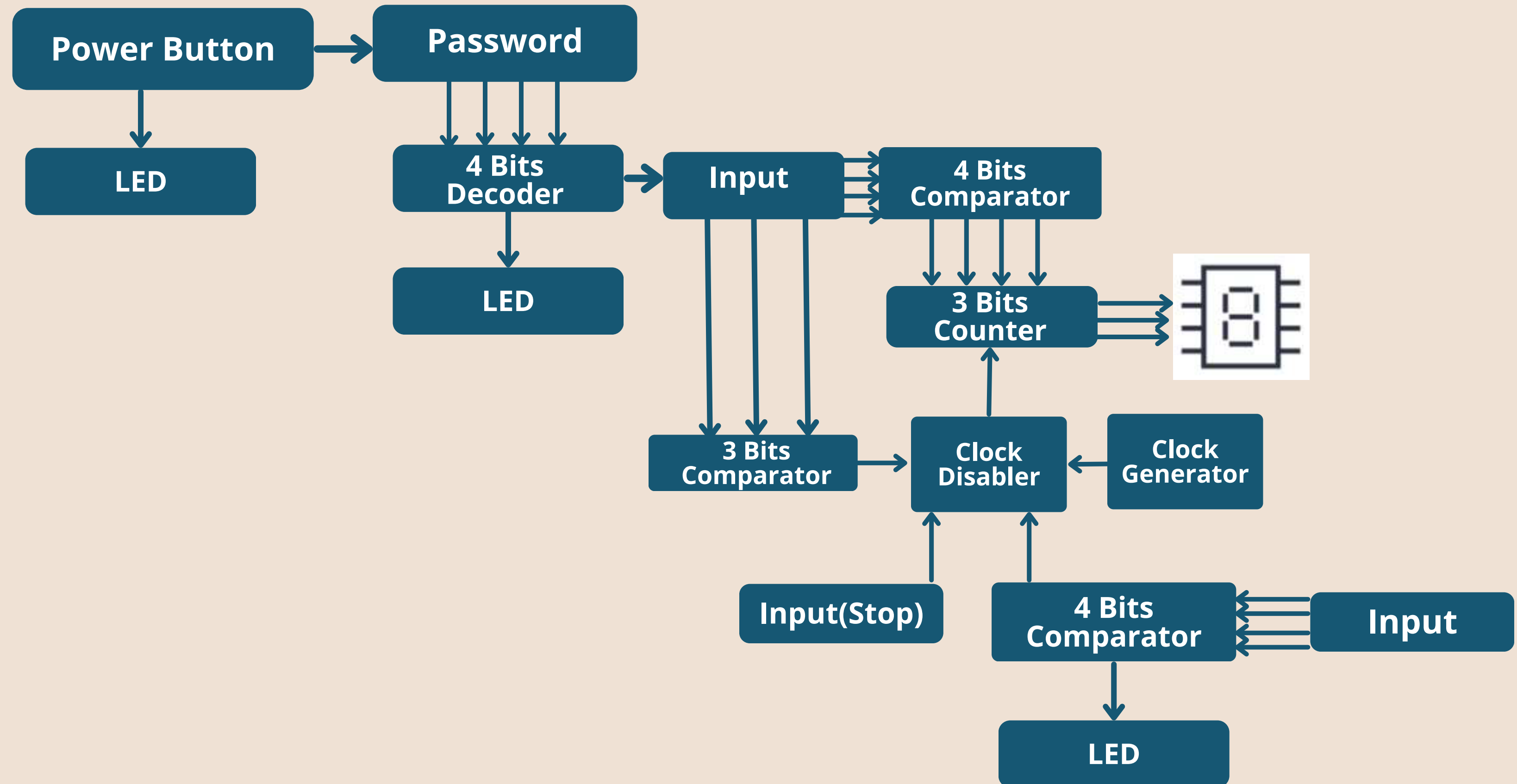
- LED lights
- 4-digit password
- emergency stop
- maximum weight limit

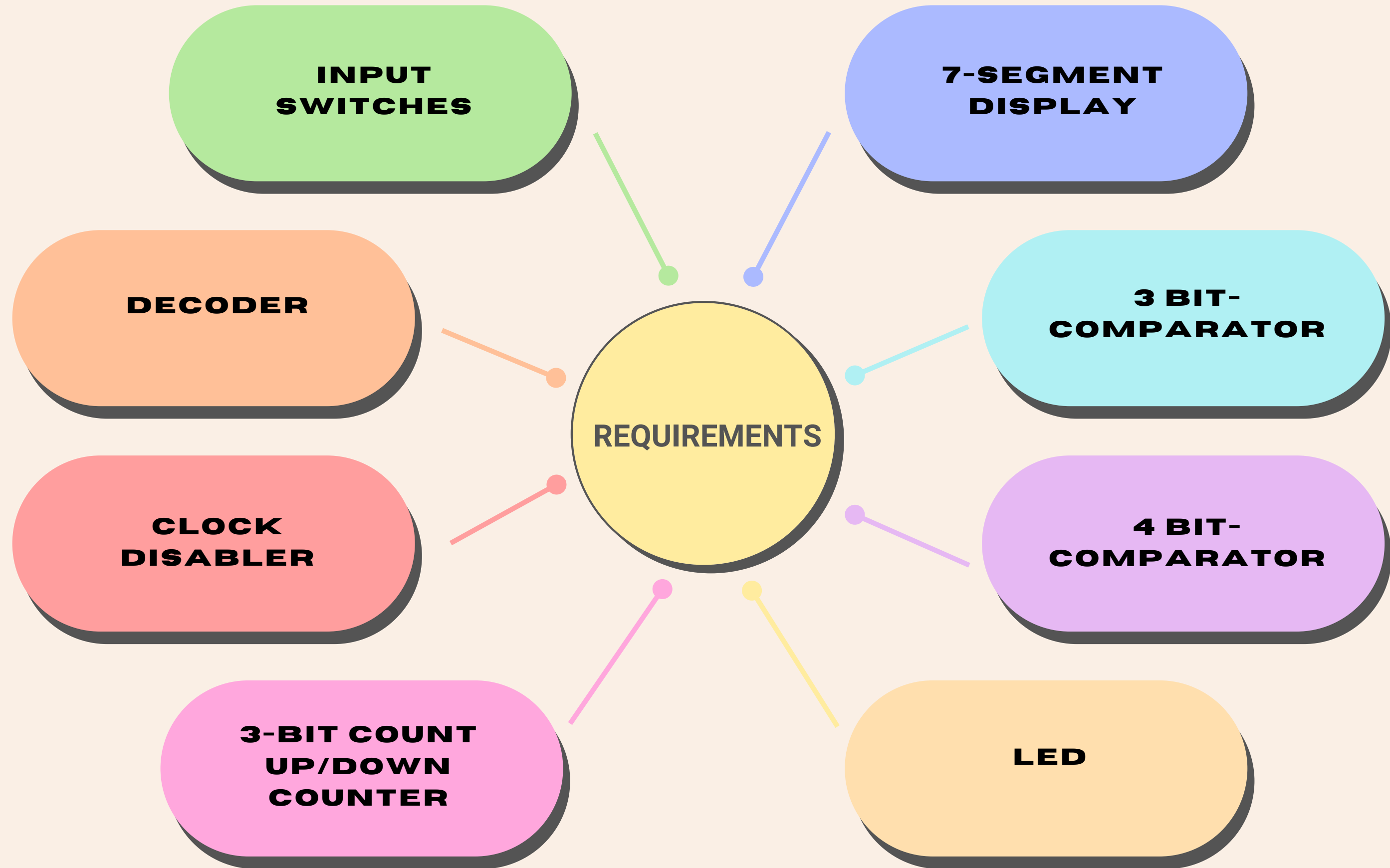
Problem & Solutions

- **Precision in reaching the intended destination**
A circuit that resulted in a consistent and accurate arrival at the desired floor is implemented.
- **Ensure passenger safety**
An emergency stop feature is added to the system.
- **Prevent unauthorized access to the lift**
A password protection system is implemented.
- **Ensure lift function properly**
Maximum weight limit is added to system.



BLOCK DIAGRAM



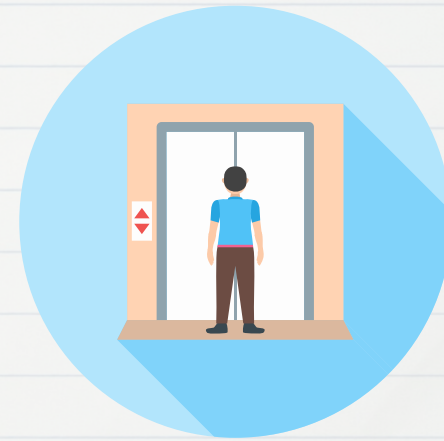
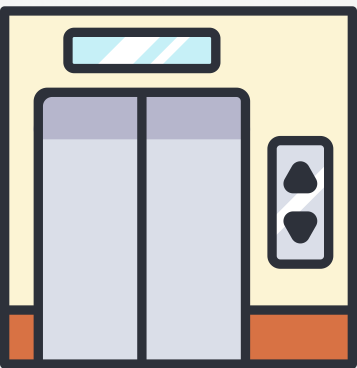




Lift Project



System Implementation



Overview of component

- Input Switch
- Decoder
- Arithmetic Comparator
- Clock Disabler
- 3-bit Up/Down Counter
- 3-bit Comparator
- 7-Segment Display



Introduction

General concepts

- Module 6 (FUNCTIONS OF COMBINATIONAL LOGIC)
- Module 7 (LATCHES AND FLIP-FLOPS)
- Module 8 (COUNTERS)

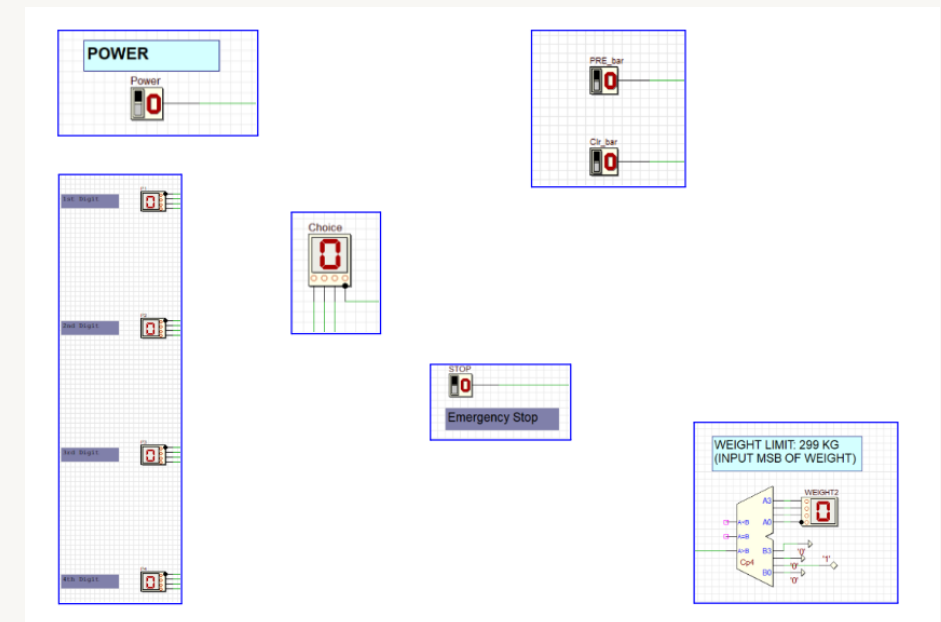
The Design Tool

Deeds



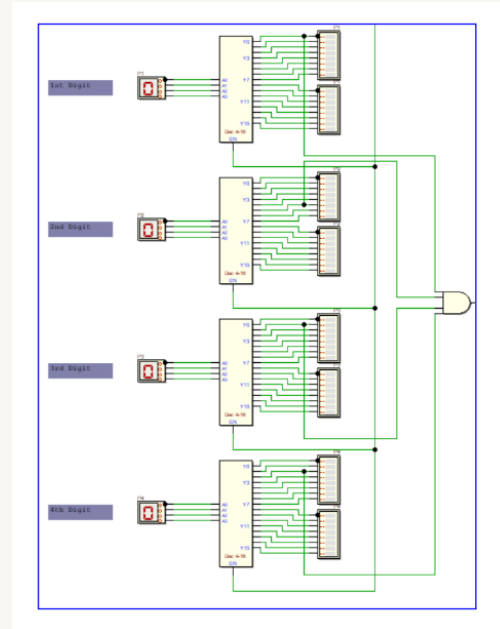
Input Switch

- Power Control
- Password Enter
- Preset/Clear Control
- Weight Sensor
- User Input for Chosen Floor



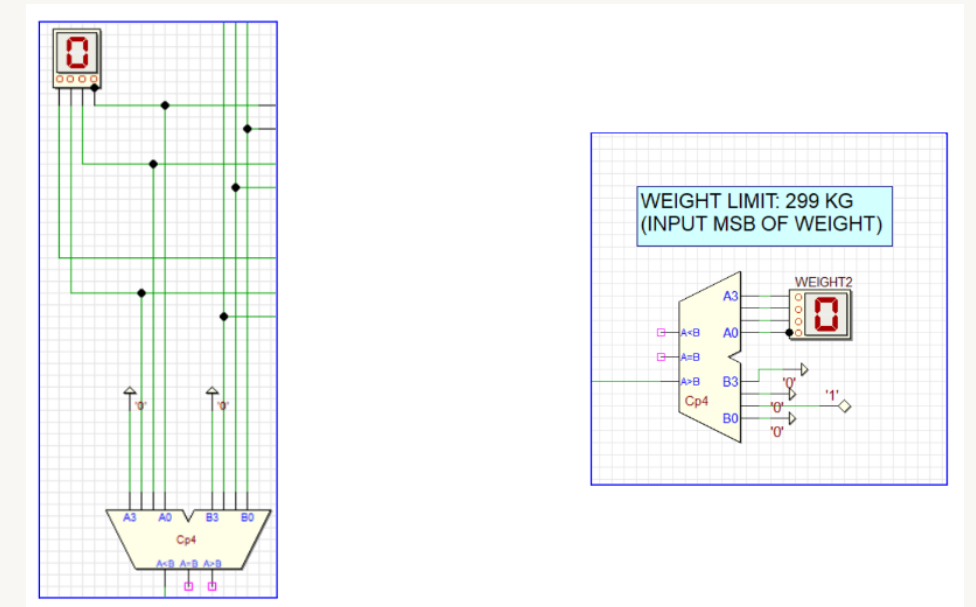
Decoder

- Connected from Power Button and Hex Digit input switch
- Each decoder connect to 2 LED array to represent the value of user key in.



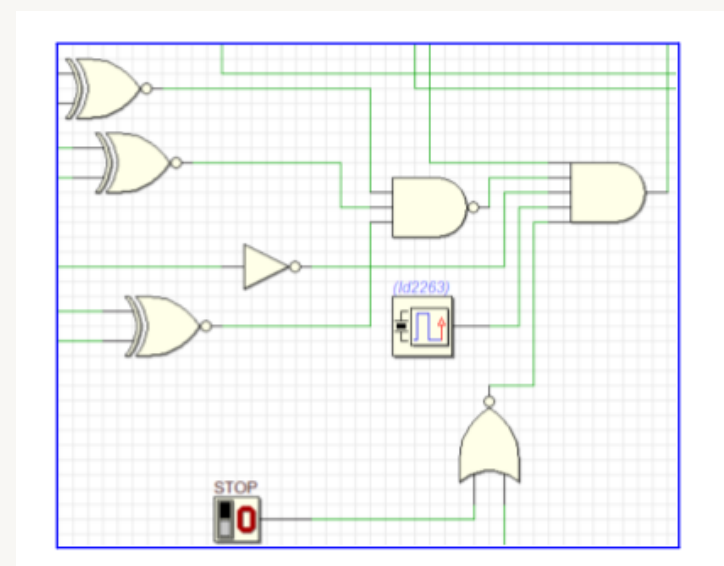
Arithmetic Comparator

- Compare the current floor with user chosen floor
- Compare the weight detect with the weight limitation



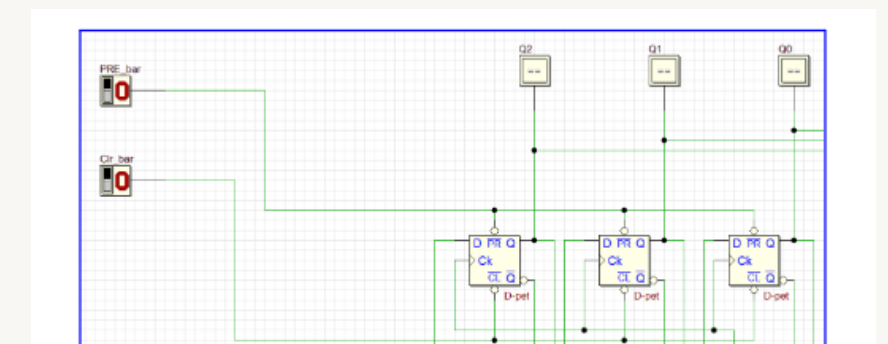
Clock Disabler

- Clock Source
- Password Detector
- Comparator
- Input Validator
- Stop/Overweight signal



3-bit up/down counter

- Start/Stop depend Clock disabler
- Connected with Pre_bar & Clr_bar
- 0 = count up
- 1 = count down
- Connect to 7-Segment Display

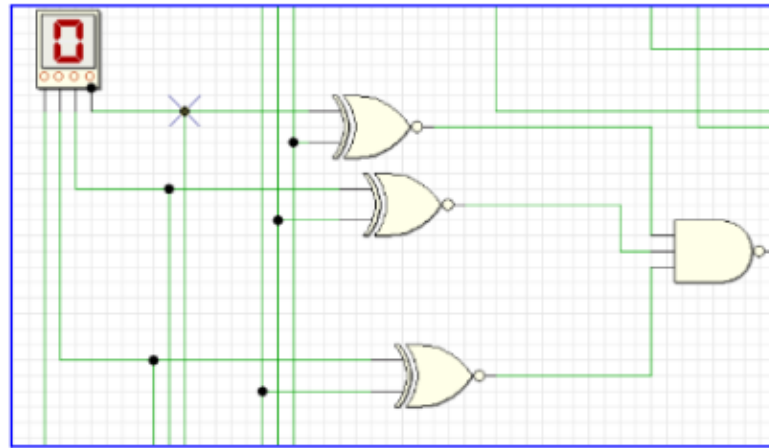


Q_2	Q_1, Q_0			
	00	01	11	10
00			1	
01	1	1		
11		1	1	1
10	1			

$$D_2 = \bar{X} \bar{Q}_2 \bar{Q}_1 + X \bar{Q}_2 Q_1 + Q_2 Q_1 \bar{Q}_0 + X \bar{Q}_2 \bar{Q}_1 Q_0 + \bar{X} \bar{Q}_2 Q_1 Q_0$$

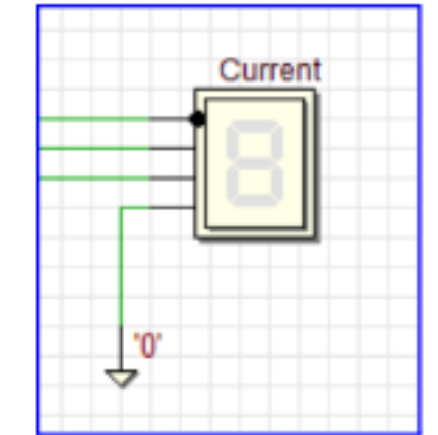
3-bit Comparator

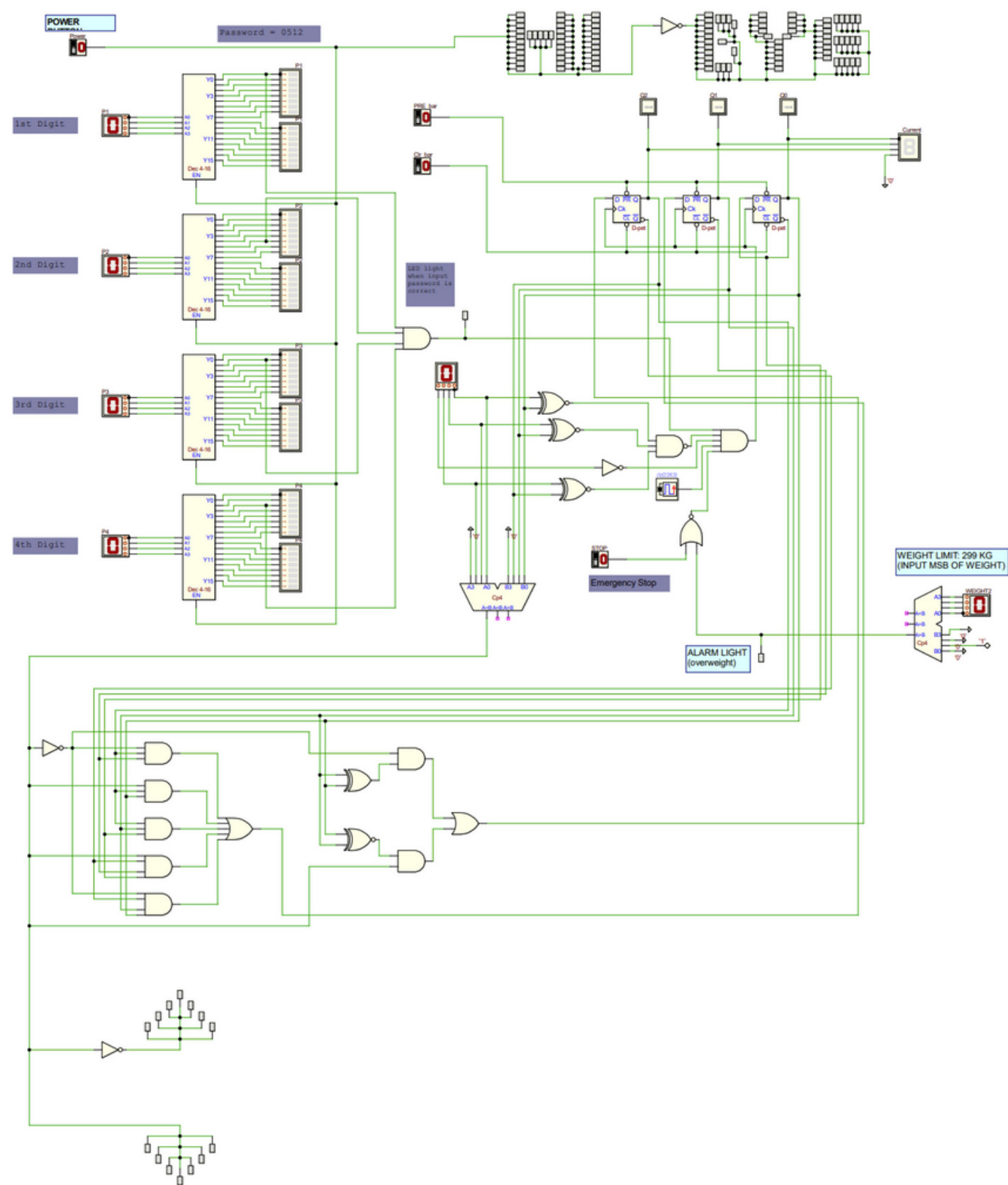
- Compare current floor with user chosen floor
- Each XNOR compare 1 digit
- XNOR show 1 when digit equal
- NAND show 0 when value equal



7-Segment Display

- Connect from the output of D FF
- Show value range 0-7





FULL DEED DRAWING

The integration of advanced features, including a 4-digit password, emergency stop, and weight sensor, reflects a commitment to addressing critical aspects such as security, safety, and operational precision. The proficiency demonstrated in circuit design, utilizing combinational and sequential circuits with various components, showcases a strong understanding of digital logic principles.

conclusion

our achievements have been notable in several key areas. The successful collaboration within the team stands out as a significant accomplishment, as each member brought unique skills, dedication, and perspectives, contributing to the efficient completion of the project

ENSURING A CONSISTENT AND ACCURATE ARRIVAL AT THE DESIRED FLOOR, THEREBY ENHANCING THE OVERALL USER EXPERIENCE.

PRIORITIZING SAFETY THROUGH FEATURES LIKE THE EMERGENCY STOP AND PASSWORD PROTECTION SYSTEM HIGHLIGHTS A COMMITMENT TO PASSENGER WELL-BEING AND SECURITY. THE COMPREHENSIVE DESIGN OF THE LIFT ELECTRONIC CONTROLLER SYSTEM, MEETING BASIC FUNCTIONALITY REQUIREMENTS WHILE INCORPORATING ADDITIONAL FEATURES, IS ANOTHER NOTABLE STRENGTH.

WEAKNESS

CLEARER AND BETTER ORGANIZED TO ENHANCE OVERALL READABILITY AND UNDERSTANDING

REPORT BRIEFLY MENTIONS FUTURE WORK WITHOUT PROVIDING SPECIFIC DETAILS ON AREAS FOR IMPROVEMENT OR EXPANSION, INDICATING A NEED FOR MORE DETAILED PLANNING AND PROPOSALS FOR LONG-TERM ENHANCEMENTS.

FUTURE WORK COULD INCLUDE REFINING THE USER INTERFACE FOR BETTER INTERACTION, EXPLORING SYSTEM SCALABILITY, INCORPORATING ACCESSIBILITY FEATURES, AND EVALUATING ENERGY EFFICIENCY MEASURES. BY LEVERAGING OUR STRENGTHS AND ADDRESSING WEAKNESSES, WE AIM TO CONTINUALLY IMPROVE THE LIFT ELECTRONIC CONTROLLER SYSTEM, ENSURING ITS EFFECTIVENESS, SECURITY, AND ADAPTABILITY IN THE LONG RUN

THANK FOR WATCHING